

Cell Type Bacterial, gram negative

Molecules Electroporated DNA, various supercoiled plasmids - CsCl pure; Qigen column, after microdialysis

Species Used *E. coli*, N99, DH 5 α

Before the Pulse

Cell Growth Medium LB

Growth Phase at Harvest O.D. (600) = whatever

Pre-pulse Incubation Short as possible; however long it takes to set up.

Wash Solution Twice in cold water; twice in 10% glycerol

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature Not given

Electroporation Medium 10% glycerol

Cuvette Gap 0.2 cm

Voltage 2.5 kV

Cell Density Not given

Volume of Cells Not given

Field Strength 12.5 kV/cm

DNA Concentration Not given

DNA Resuspension Buffer TE (10 mM Tris, 1 mM EDTA, pH 8.0)

Capacitor 25 μ F

Volume of DNA 1 to 10 μ l

Resistor 200 Ω (Pulse Controller)

After the Pulse

Time Constant 4.0 to 5.0 msec

Outgrowth Medium SOC (Add glucose after autoclaving and cooling)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses. I have electroporated temperature-sensitive mu-lysogens with no problems. I was concerned, because even a slight increase in temperature will result in cell death. As long as I keep things on ice and work quickly, there seems to be no problem with temperature induction by the electroporation.

SOC: 2% Bacto tryptone, 0.5% Bacto yeast extract, 10mM NaCl, 2.5mM KCl, 10 mM MgCl₂, 10 mM MgSO₄, 20 mM glucose.

LB: 1% Bacto tryptone, 0.5% Bacto yeast extract, 0.5% NaCl.

Outgrowth Temperature 30 °C (N99) 37 °C (DH5a)

Length of Incubation 4hr (N99); 2hr (DH5a)

Selection Method or Assay Used Ampicillin, kanamycin resistance

Electroporation Efficiency 2 to 4 x 10⁸ (8) transformants / μ g DNA

Per Cent Survival Not given

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041